

Trishna Saikia

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Google Scholar

LinkedIn

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Personal Homepage

EDUCATION

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| • Indian Institute of Technology | Indore, India |
| • <i>Doctor of Philosophy - Computer Science and Engineering ; CPI: 8.5/10</i> | <i>Aug 2022 - Oct 2026 (Expected)</i> |
| • National Institute of Technology | Jamshedpur, India |
| • <i>Master of Technology - Computer Science and Engineering; CPI: 9.18/10</i> | <i>Aug 2020 - Jul 2022</i> |
| • Gauhati University Institute of Science and Technology | Guwahati, India |
| • <i>Bachelor of Technology - Information Technology; CPI: 9.02/10</i> | <i>Aug 2015 - Jul 2019</i> |

RESEARCH PROFILE

I am a Ph.D. candidate in Computer Science and Engineering at the Indian Institute of Technology Indore, specializing in multimodal AI, computer vision, and physiological computing. My research focuses on developing reliable and adaptive machine-learning methods for extracting physiological information from facial videos and heterogeneous biomedical signals.

My work spans contactless estimation of heart rate, respiratory rate, blood pressure, blood oxygen saturation, and hemoglobin, along with multimodal pain assessment and physiology-based deepfake detection. My methodological interests include cross-modal supervision, knowledge distillation, temporal modeling, uncertainty-aware learning, signal-quality assessment, and robust evaluation across occlusion, video compression, demographic variation, and cross-dataset settings.

I also have extensive experience in human-subject data collection, physiological dataset development, model validation, and interdisciplinary research. My long-term goal is to develop personalized, trustworthy, and deployable human-AI systems that integrate multimodal spatiotemporal data for healthcare and rehabilitation.

RESEARCH EXPERIENCE

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| • Multimodal and Trustworthy AI for Physiological Computing | <i>July 2022 - Present</i> |
| ◦ Developed deep learning methods for non-contact physiological monitoring from facial videos. | |
| ◦ Applied AI-based physiological sensing to respiratory rate, blood pressure, SpO ₂ , and heart-rate estimation. | |
| ◦ Worked on healthcare-related dataset collection, preprocessing, and robust model evaluation. | |
| ◦ Investigated uncertainty, quality-aware, robustness-oriented learning for heterogeneous and noisy physiological data. | |
| ◦ Worked with real-world acquisition challenges including motion, occlusion, compression, and cross-subject variability. | |
| • Physiological-Signal Reasoning for Deepfake and Affective Computing | <i>July 2022 - July 2025</i> |
| ◦ Investigated physiological inconsistencies in facial videos for deepfake detection by using remote photoplethysmography. | |
| ◦ Worked on pain assessment and neonatal pain detection using physiological and multimodal cues. | |
| • Medical Imaging and Computational Healthcare AI | <i>July 2020 - Dec 2023</i> |
| ◦ Developed deep learning methods for lung nodule classification using CT DICOM images. | |
| ◦ Worked on cancer diagnosis and prognosis-related imaging analysis. | |
| ◦ Worked on medical image preprocessing, feature extraction, transfer learning, and clinical AI evaluation. | |

JOURNAL ARTICLES

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- **Trishna Saikia**, Puneet Gupta, and Pasi Liljeberg.: “BP-rPPG: An Indian Face-Video Dataset and PPG-Guided Baseline for Remote Blood Pressure Estimation,” *IEEE Transactions on Consumer Electronics*, 2026. DOI: <https://doi.org/10.1109/TCE.2026.3719074>.
 - **Trishna Saikia**, Satwik Vankayalapati, Puneet Gupta, and Pasi Liljeberg.: “VOLEMIA: Non-Invasive Blood Pressure Estimation Using Temporal-Spectral Convolutional Network,” *Digital Signal Processing*, 2025. DOI: <https://doi.org/10.1016/j.dsp.2025.105393>.
 - Vaidehi Agarwal, **Trishna Saikia**, Anup Kumar Gupta, and Puneet Gupta.: “SHINE: Synergizing Transformers with Contrastive Learning for Thriving rPPG-based SpO₂ Estimation,” *Expert Systems with Applications*, 2025. DOI: <https://doi.org/10.1016/j.eswa.2025.129190>.
 - Lokendra Birla, **Trishna Saikia**, and Puneet Gupta.: “AVENUE: A Novel Deepfake Detection Method Based on Temporal Convolutional Network and rPPG Information,” *ACM Transactions on Intelligent Systems and Technology*, 2025. DOI: <https://doi.org/10.1145/3702232>.
 - **Trishna Saikia**, Lokendra Birla, Anup Kumar Gupta, and Puneet Gupta.: “HREADAI: Heart Rate Estimation from Face Mask Videos by Consolidating Eulerian and Lagrangian Approaches,” *IEEE Transactions on Instrumentation and Measurement*, 2023. DOI: 10.1109/TIM.2023.3334359.
 - **Trishna Saikia**, Ranjeet Kumar, Dilip Kumar, and Koushendra Kumar Singh.: “An Automatic Lung Nodule Classification System Based on Hybrid Transfer Learning Approach,” *SN Computer Science*, 2022. DOI: <https://doi.org/10.1007/s42979-022-01167-0>.

CONFERENCE ARTICLES

- **Trishna Saikia**, Anup Kumar Gupta, Puneet Gupta, and Pasi Liljeberg.: “Exploring PPG-guided Knowledge Distillation for Contactless Respiration Estimation,” *IEEE International Conference on Automatic Face and Gesture Recognition*, 2026. DOI: <https://doi.org/10.1109/FG67764.2026.11557053>.
- Anup Kumar Gupta, **Trishna Saikia**, Puneet Gupta, and Abhinav Dhall.: “PainXtract: A Multimodal System for Multiclass Pain Classification Using Physiological Signals,” *International Conference on Multimodal Interaction*, 2025. DOI: <https://doi.org/10.1145/3747327.3764792>.
- **Trishna Saikia**, Ashutosh Dhamaniya, Puneet Gupta, and Koushlendra Kumar Singh.: “An Ensemble Technique for Classification of Oral Cancer by Using Histopathological Imaging,” In *International Conference on Machine Vision and Augmented Intelligence* (2023), DOI: 10.1007/978-981-97-4359-9_66
- Lokendra Birla, Sneha Shukla, **Trishna Saikia**, and Puneet Gupta.: “HR-TRACK: An rPPG Method for Heart Rate Monitoring Using Temporal Convolution Networks,” *International Conference on Pattern Recognition*, 2024. DOI: https://doi.org/10.1007/978-3-031-78201-5_24.
- **Trishna Saikia**, Ashutosh Dhamaniya, Puneet Gupta, and Koushlendra Kumar Singh.: “An Ensemble Technique for Classification of Oral Cancer by Using Histopathological Imaging,” *International Conference on Machine Vision and Augmented Intelligence*, 2023. DOI: https://doi.org/10.1007/978-981-97-4359-9_66.
- **Trishna Saikia**, Malho Hansdah, Koushlendra Kumar Singh, and Manish Kumar Bajpai.: “Classification of Lung Nodules Based on Transfer Learning with K-Nearest Neighbor (KNN),” *Imaging System and Technology*, 2022. DOI: <https://doi.org/10.1109/IST55454.2022.9827668>.
- **Trishna Saikia** and Koushlendra Kumar Singh.: “Computer-Aided Leaf Disease Classification Based on Logistic Regression and Transfer Learning Approach,” *International Conference on Machine Vision and Augmented Intelligence*, 2022. DOI: https://doi.org/10.1007/978-981-99-0189-0_33.

SUBMITTED / UNDER REVIEW

- **Trishna Saikia**, Anup Kumar Gupta, Puneet Gupta, and Pasi Liljeberg.: “Respiratory Rate Estimation via Cross-Modality Supervision of Lagrangian and Eulerian rPPG with Skin-Tone and Illumination Awareness,” *ACM Transactions on Computing for Healthcare*, Under Review.
- Lokendra Birla, **Trishna Saikia**, Puneet Gupta, and Pasi Liljeberg.: “Uncertainty-Aware Multi-Task Learning for Joint rPPG and Hemoglobin Estimation,” *IEEE Transactions on Artificial Intelligence*, Under Review.
- Vipendra Singh, **Trishna Saikia**, Anup Kumar Gupta, and Puneet Gupta.: “Can Real and Synthetic Face Videos Be Physiologically Modulated? A Training-Free Inverse rPPG Approach,” *IEEE Signal Processing Letters*, Under Review.
- Debashish Utpal Biswas, **Trishna Saikia**, Anup Kumar Gupta, and Puneet Gupta.: “CFOCUS: A Contrastive Transformer Method for Ocular Driver Drowsiness Detection with Subject-Specific Calibration,” *Machine Vision and Applications*, Submitted.

PREPRINTS

- Ashutosh Dhamaniya, Anup Kumar Gupta, **Trishna Saikia**, and Puneet Gupta.: “Exploring Remote Photoplethysmography for Neonatal Pain Detection from Facial Videos,” arXiv preprint arXiv:2604.25680, 2026.

BOOK CHAPTER

- **Trishna Saikia** and Koushlendra Kumar Singh.: “Introduction to Various Optimization Techniques,” in *Nature-Inspired Optimization Algorithms and Soft Computing*, IET Book, Chapter 1. ISBN: 978-1-83953-517-8.

PATENTS

- Puneet Gupta, **Trishna Saikia**, Anup Kumar Gupta.: “System, apparatus, and method for heart rate determination of individuals wearing face mask”, Under filing in Indian jurisdiction (Ref. No: 202321071568).

DATASETS WITH BASELINES

- **BP-rPPG Dataset and ALIVE Baseline.**: An Indian face-video dataset and PPG-guided baseline for remote blood pressure estimation.
<https://github.com/TrishnaSaikia/BP-rPPG-ALIVE>
- **DIL-RR Dataset and Baseline.**: A facial-video dataset and baseline for contactless respiratory rate estimation.
<https://github.com/TrishnaSaikia/DIL-RR>
- **DIL-rPPG Dataset and HREADAI Baseline.**: A facial-video dataset and baseline for heart rate estimation from face-mask videos.
<https://github.com/TrishnaSaikia/DILrPPG-Dataset-and-HREADAI-Baseline>

PROJECTS

- **Vehicle Tracking and Crowdedness Detection Inside the Vehicle** Jan. 2019 - Jul. 2019
 - Developed a real-time bus tracking and crowdedness estimation system.
 - Worked on video/image-based analysis, detection, and practical deployment-oriented system design.
 - Relevant skills: computer vision, tracking, OpenCV, machine learning, real-time sensing.
- **Smart Face Recognizing Locking System** Sep. 2018
 - Developed an intelligent face-recognizing locking system at Utkranti 1.0, organized at GUIST, Guwahati.
 - Framework: OpenCV and Arduino.
- **ChatBot on NorthEast tourism** Apr. 2018
 - Developed a chatbot on NorthEast tourism, the Data Science Hackathon organized at IIT Guwahati.
 - http://t.me/sangrakt_bot.

TEACHING AND MENTORING EXPERIENCE

- **Guest Faculty under PMRF Scheme, GUIST** *Jan. 2025 – Jul. 2026*
 - Taught System Programming, Web Technology, and Python Programming.
- **Guest Faculty under PMRF Scheme, SGSITS Indore** *Mar. 2023 – Dec. 2024*
 - Taught Deep Learning and Reinforcement Learning.
- **Teaching Assistant, IIT Indore** *Aug. 2022 – Present*
 - Assisted courses: Computer Vision, Pattern Recognition, Software Engineering, and Data Structures and Algorithms.
 - Mentored 5 Master's students and 2 junior PhD scholars at the Deep Intelligence Lab, IIT Indore.

INTERNATIONAL RESEARCH EXPERIENCE

- **Visiting Research Scholar, University of Turku, Finland** *Apr. 2026 – Jun. 2026*
 - Erasmus+ funded research stay at the Department of Computing, University of Turku.
 - Worked under the supervision of Prof. Pasi Liljeberg on AI and physiological computing research.
 - Gained experience in an international and interdisciplinary Finnish research environment.

TECHNICAL SKILLS & ACHIEVEMENTS

Technical Skills:

- **Programming:** Python, MATLAB, C
- **Machine Learning:** Classification, regression, clustering, dimensionality reduction, and model evaluation
- **Deep Learning:** PyTorch, TensorFlow, CNNs, RNNs, LSTMs, Transformers, and TCN
- **Computer Vision:** OpenCV, video processing, object/face detection, ROI extraction, visual feature extraction
- **Signal and Sensor Data Processing:** filtering, FFT, PSD analysis, feature extraction, sensor/video signal analysis
- **Data Handling and Analysis:** CSV/Excel data processing, NumPy, pandas, scikit-learn
- **Tools:** LaTeX, Git, SageMath, Microsoft Excel

Achievements:

- Selected as an India RISE Fellow (2026).
- Selected for an Erasmus+ funded visiting research stay at the University of Turku, Finland, April–June 2026.
- Institute Best Technology Award, 2025, Indian Institute of Technology (IIT) Indore.
- Prime Minister Research Fellowship scholar, IIT Indore.
- Received incubation support from Assam Startup Incubation facilities during B.Tech.
- Received Ishan Uday Fellowship, in B.Tech from the Department of IT, Gauhati University (2015-19 batch)
- 1st position in B.Tech from the Department of IT, Gauhati University (GU) (2015-19 batch).

OTHER ACTIVITIES:

- Served as a peer reviewer for International Conferences like ICASSP, FG, NeurIPS, ECCV and journals including: IEEE Transactions on Affective Computing (TAFEC), IEEE Transactions on Systems, Man, and Cybernetics: Systems, Biomedical Signal Processing and Control, IEEE Transactions on Instrumentation and Measurement, Expert Systems with Applications, Pattern Recognition, Engineering Applications of Artificial Intelligence, Signal, Image and Video Processing, and SN Computer Science.
- Contributed to research documentation: project reports, and proposal writing for interdisciplinary AI and healthcare research.
- Research Mentoring: Mentored 3 Master's students and 2 junior PhD scholars at the Deep Intelligence Lab, IIT Indore.

LANGUAGE SKILLS

- **English:** Professional working proficiency
- **Hindi:** Fluent
- **Assamese:** Native

REFERENCES

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